

**JUSTIFICATION AND APPROVAL
FOR OTHER THAN FULL AND OPEN COMPETITION
SPRDL1-23-R-0099**

1. Contracting Agency: Defense Logistics Agency - DLA Land Warren (DLA-WRN).

2. Description of Action: DLA-WRN proposes to award a new Firm-Fixed-Price contract to DRS Network & Imaging Systems, LLC (DRS), Commercial and Government Entity (CAGE) 24290, for National Stock Number (NSN) 6150-01-699-8530, using Army Working Capital Funds.

3. Description of Supplies/Services: The NSN 6150-01-699-8530, vendor part number PR63636 / Army Ordinance Part Number (AOPN) 12594593, Electrical Distributing System, is used on the Armored Multi-purpose Vehicle (AMPV). The deliverables for this contract are for a five-year long-term Indefinite Delivery Indefinite Quantity (IDIQ) contract with a guaranteed minimum quantity of 68 each and a total maximum quantity of 325 each. The total estimated maximum value of this contract is \$ [REDACTED]

The Distributing System, Electrical, is the Power Management Diagnostic Controller (PMDC) located on the left sponson next to the commander's station. This item distributes electrical power to the vehicle, and it stores information for diagnostics of vehicle electrical systems and components. The next higher assembly is the Common Vetronics installation, and the end item is the AMPV.

Acquisition Method Code (AMC):

4 - Acquire, for the first time, directly from the actual manufacturer.

Acquisition Method Suffix Code (AMSC):

B – This part must be acquired from a manufacturing source(s) specified on a source control or selected item drawing as defined by the current version of DoD-STD-100. Suitable technical data, Government data rights, or manufacturing knowledge are not available to permit acquisition from other sources, nor qualification testing of another part, nor use of a second source part in the intended application. Although, by DoD-STD-100 definition, altered and selected items shall have an adequate technical data package, data review discloses that required data or data rights are not in government possession and cannot be economically obtained. If one source is available, AMCs 3, 4 or 5 are valid. If at least two sources exist, AMCs 1 or 2 are valid.

4. Authority Cited: The statutory authority permitting other than full and open competition for this effort is 10 United States Code (U.S.C.) 3204(a)(1) as implemented by the Federal Acquisition Regulation (FAR) 6.302-1, Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements.

5. Reason for Authority Cited: The Distributing System, Electrical is a power distribution, video

processing, signal conditioning, process control, and embedded diagnostics Line Replacement Unit (LRU). It monitors discrete inputs and sensors, and distributes electrical power to control AMPV internal/external lighting, fuel pumps, ramp, personnel heater, bilge pumps, horn, etc. It supplies the instrumentation and video seen on the AMPV Driver display. The power unit is key to the vehicle's electronics.

The Government does not have the pertinent technical data to enable competitive re-procurement. The Government does not own the necessary intellectual property data rights to the pertinent technical data to enable competitive re-procurement.

The information needed to qualify another source is defined within proprietary DRS specifications that are not in the Government's possession. The Distributing System, Electrical, is defined by DRS Performance Requirement, PE53417, and is tested in accordance with Acceptance Test Procedure PE53448. Environmental Stress Screening (ESS) must be performed in accordance with PE53475. All three specification are reference within AOPN drawing 12594593 but were not delivered to the Government as a portion of the Technical Data Package (TDP). DRS was not funded to deliver a complete TDP of the Distributing System, Electrical, and there are currently no plans to acquire a competitive TDP.

Due to the lack technical data available for the Distributing System, Electrical, Reverse Engineering (RE) is the only recourse for the Government to increase competition through development of a competitive build-to-print TDP. A significant amount of time and money would be necessary to reverse engineer, procure and test prototypes, and release the design. Through our previous experience, we estimate that the cost to develop a competitive procurement package for the Distributing System, Electrical, is estimated at \$3,973,080 to include fabrication and testing of the prototype hardware and releasing a validated TDP. Vehicle level qualification and durability testing would be required to prove out this item if manufactured by a new source at an estimated cost of \$1.28M. Fabrication of reverse engineered prototypes would cost up to \$940,000.

Table 1. Reverse Engineering Breakdown	
Development and Hardware Design	\$90,000
Develop and Validate TDP	\$940,000
RAM Testing	\$1,280,000
Nuclear Hardness	\$723,080
Integration & Fabrication/Installation	\$940,000
Total:	\$3,973,080

The Government would not be expected to recover in a reasonable amount of time, through subsequent competition, the duplication of costs involved in reverse engineering the Distributing System, Electrical, to create a new and competitive TDP. At [REDACTED] the current unit cost of \$ [REDACTED], and assuming 25% savings achieved through competition, the \$ [REDACTED] in annual savings would take over [REDACTED] years to offset the estimated \$3.973M cost of reverse engineering the item.

Calculation:

Payback = Cost to reverse Engineer / Annual savings
= Cost to [REDACTED] / (Annual Demand * unit price * 25% savings)
= [REDACTED] * 0.25)
= \$3,973,0 [REDACTED]
= [REDACTED] Years

6. Efforts to Obtain Competition:

- a. Effective competition: Due to the reasons cited in section 5 above we are not pursuing competition for this item.
- b. Subcontracting competition: Unless the contract is awarded to a small business, the solicitation and resulting contract will include FAR clauses 52.219-8, "Utilization of Small Business Concerns"; 52.244-5, "Competition in Subcontracting"; 52.219-16, "Liquidated Damages--Subcontracting Plan"; and Defense Federal Acquisition Regulation Supplement (DFARS) 252.219-7003, "Small Business Subcontracting Plan (DoD Contracts)", to maximize small business opportunities and utilization.

SOLE SOURCE: Notices required by FAR 5.201 will be posted.

7. Actions to Increase Competition: There are no plans to increase competition for future procurements due to the reasons cited above.

Due to the costs associated with developing a competitive TDP that cannot be recovered, the Government is not actively pursuing the development of a competitive TDP for future procurements.

8. Market Research: A Sources Sought Notice was posted to SAM.gov on 3/28/2023 and was open for 30 days. There were no responses received.

Market research was done online by Mechanical engineer from Medium Combat sustainment on November 28th, 2022. It was executed as online research via sites such as Google, windchill and the Information Handling System (IHS) Markit using keywords such as " vehicle electronics power management diagnostic controller" or "vectronics+computing+unit"; and the findings pointed to the Bradley control unit used on the M2 & M3 Bradley. Unfortunately, the control unit that was found is incompatible in function to the commanders' weapon station control box. Engineering researched the existing TDP for an equivalent power control unit, and one result led to the Battery Monitoring System control unit, which cannot be used in this application as it is not the same form or function of the Distributing System, Electrical. This was done in the search within the Information Handling System (IHS) Markit and the database windchill. The findings were determined that they do not meet the same dimensional envelope as the existing Distributing System, Electrical.

Web search:

Keyword: "vehicle electronics power management diagnostic controller"
Website: http://www.actiaus.com/_wp16/solutions/multiplexing-power-management/
Finding: models do not meet fit or function of the power distribution box

Keyword; "vectronics+computing+unit"
Website: https://www.renesas.com/us/en/products/automotive-products/automotive-power-management/mcu-pmic#parametric_options
Findings: models do not meet fit or function of the power distribution box

Contacted the contractor to request consideration for the U.S. Government to purchase the intellectual property.

9. Interested Sources: To date no additional sources have expressed interest in this item.

10. Other Facts: Procurement History:

There is no procurement history available.

11. Technical Certification: I certify that the supporting data under my cognizance which is included in the justification is accurate and complete to the best of my knowledge and belief.

Name: [REDACTED]

Title: Branch Chief

Email: [REDACTED]

Signature:

7/11/2023

X [REDACTED]

Signed by: USA

12. Requirement Certification: I certify that the supporting data under my cognizance which is included in the justification is accurate and complete to the best of my knowledge and belief.

Name: [REDACTED]

Title: Branch Chief

Email: [REDACTED]

Signature:

[REDACTED]

X [REDACTED]

Signed by: [REDACTED]

13. Fair and Reasonable Cost Determination: I hereby determine that the anticipated cost to the Government for this contract action will be fair and reasonable. This

determination will be based on the results of a detailed cost and or price analysis. The contractor will not be required to submit certified cost and pricing data as the expected value of this procurement does not fall within the threshold. This item is other than commercial.

Name: Mahlmeister, Jeffrey J CIV DLA LAND AND MARITIME (USA)
Title: Contracting Officer
Email: jeffrey.mahlmeister@dla.mil
Signature:

7/10/2023

X //JJM//

Signed by: MAHLMEISTER.JEFFREY.J.1288587244

14. Contracting Officer Certification: I certify that this justification is accurate and complete to the best of my knowledge and belief.

Name: Mahlmeister, Jeffrey J CIV DLA LAND AND MARITIME (USA)
Title: Contracting Officer
Email: jeffrey.mahlmeister@dla.mil
Signature:

7/10/2023

X //JJM//

Signed by: MAHLMEISTER.JEFFREY.J.1288587244